



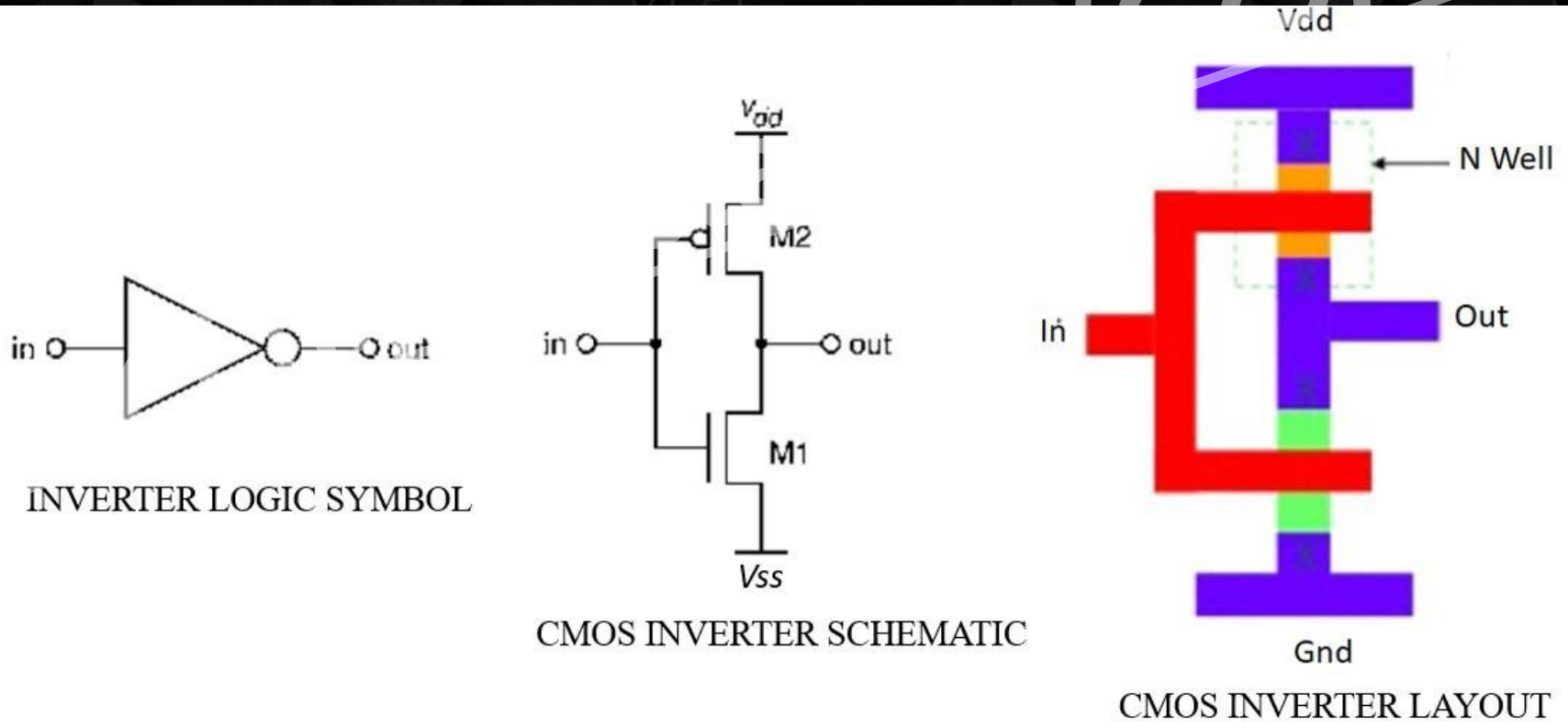
RTL2GDSII WORKSHOP DAY -1

OVERVIEW OF PHYSICAL DESIGN

Only for Chip to Startup (C2S) Training

Physical Design in VLSI

Physical design is the process of converting GATE LEVEL NETLIST (logical) to TRANSISTOR LEVEL LAYOUT (Physical) design by strictly adhering to the design rules provided by the fabrication industry.



Stages involved in Physical Design

Physical design flow consists of the following steps :-

- a. Floor planning
- b. Power planning
- c. Placement
- d. Clock Tree Synthesis
- e. Routing
- f. Timing Analysis
- g. Physical Verification

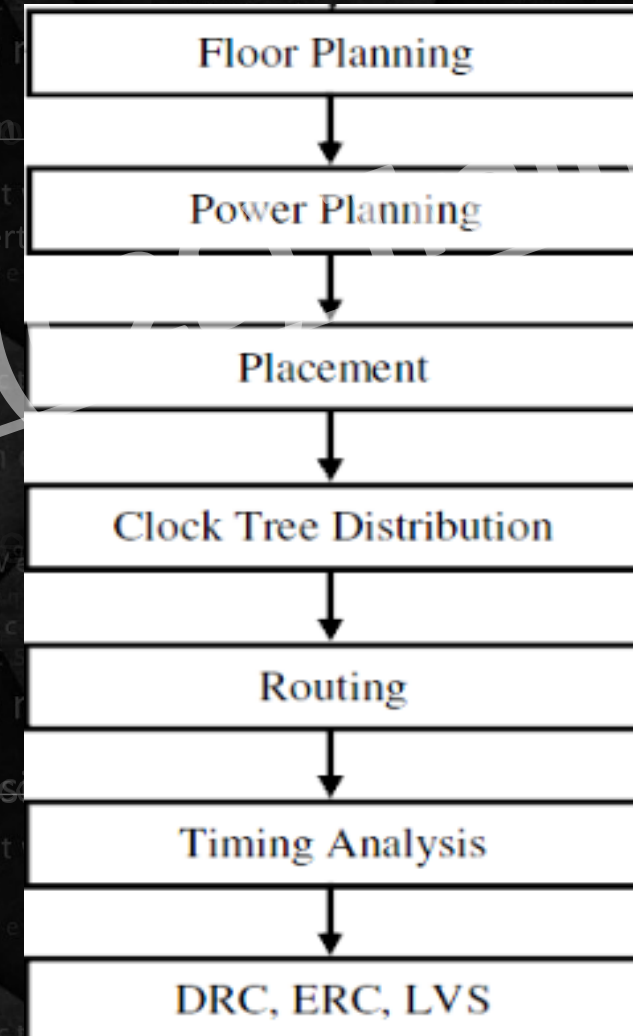


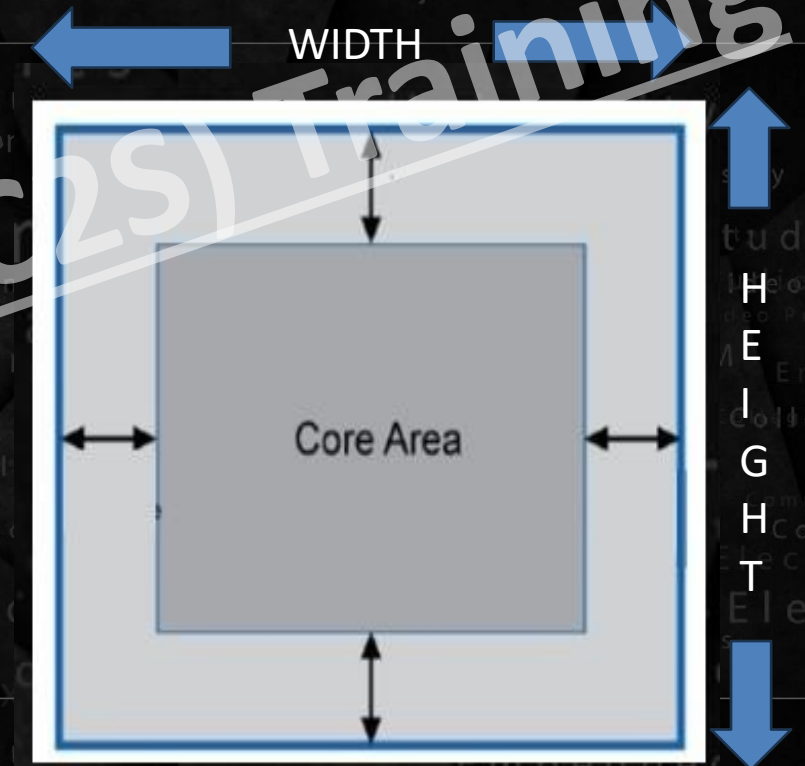
Fig. Physical Design Flow

stages involved in Physical Design.....in contrn...

a. FLOORPLANNING

Steps involved in floorplan

- define the size of chip or block (aspect ratio : W/H)
- allocates power routing resources
- place the hard macros, and
- reserve space for standard cells



DIE AREA

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stages involved in Physical Design.....in contrn...

b. Power planning

creating power network to distribute the power uniformly across chip

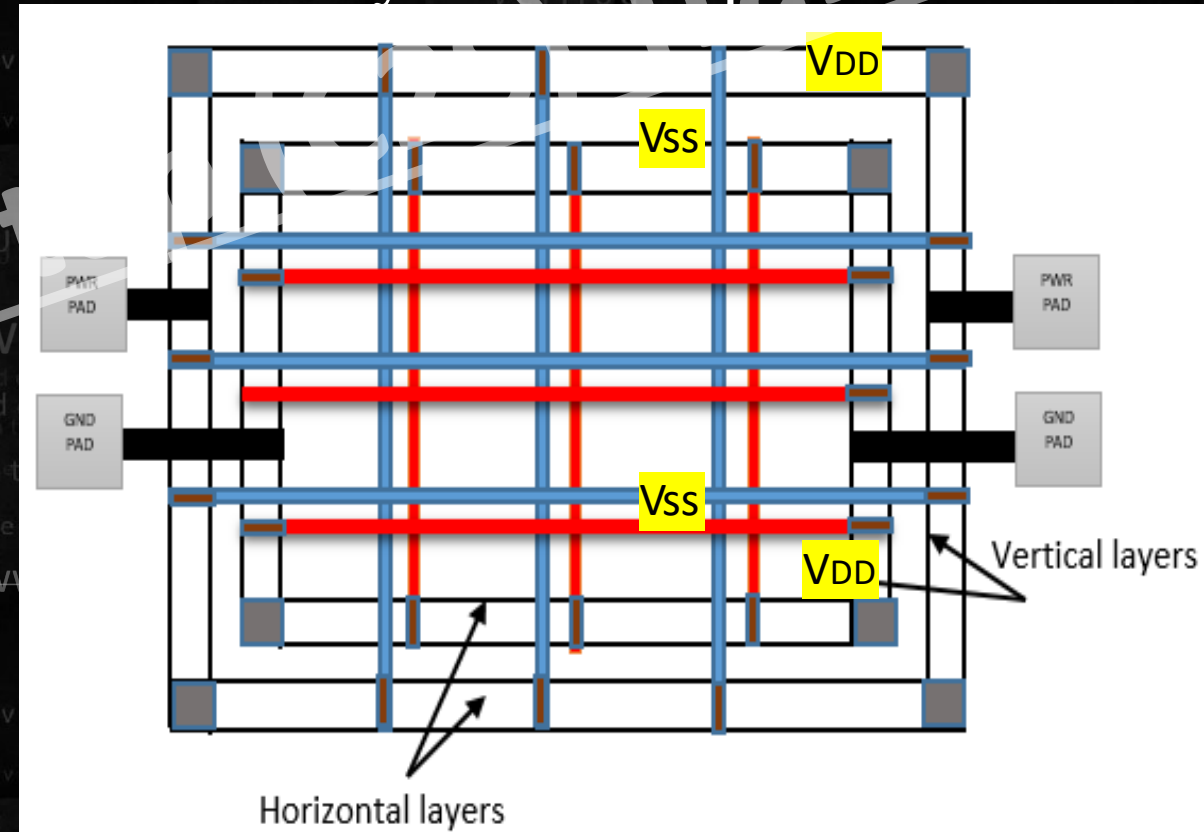
• Steps involved in power planning

Create Power ring (VDD and VSS nets)

Create Power Mesh (Power Straps)

Define VDD and VSS rails

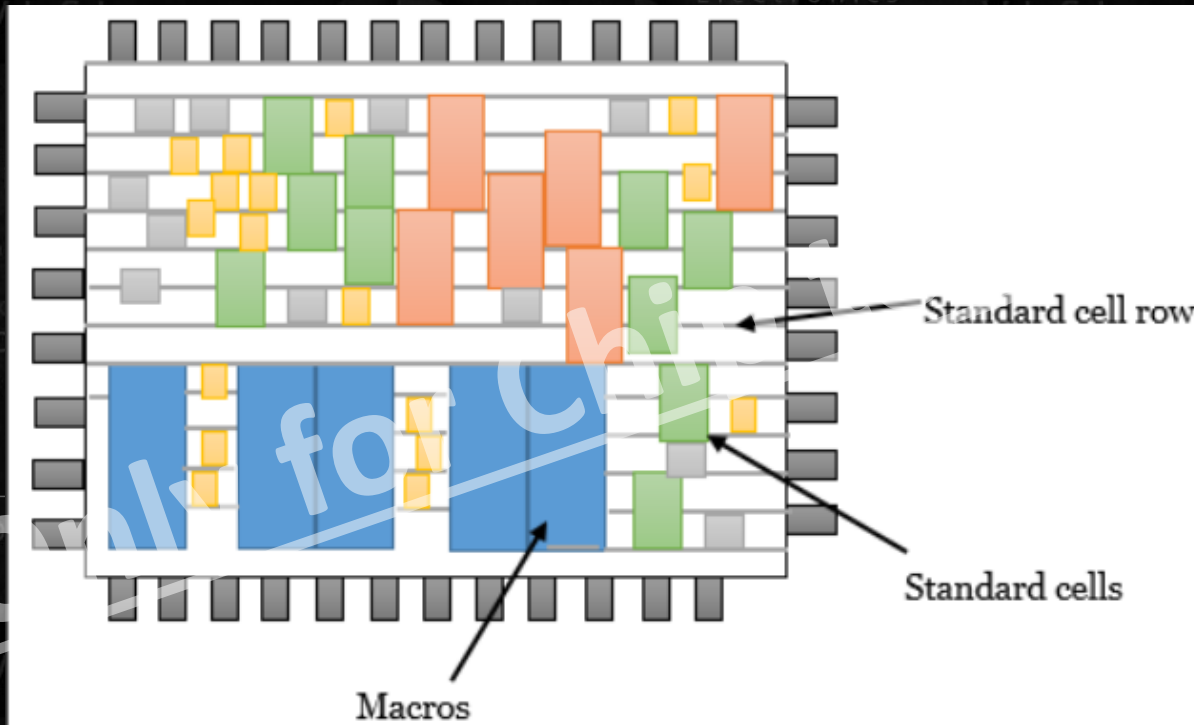
MACROS – SEPARATE POWER RING



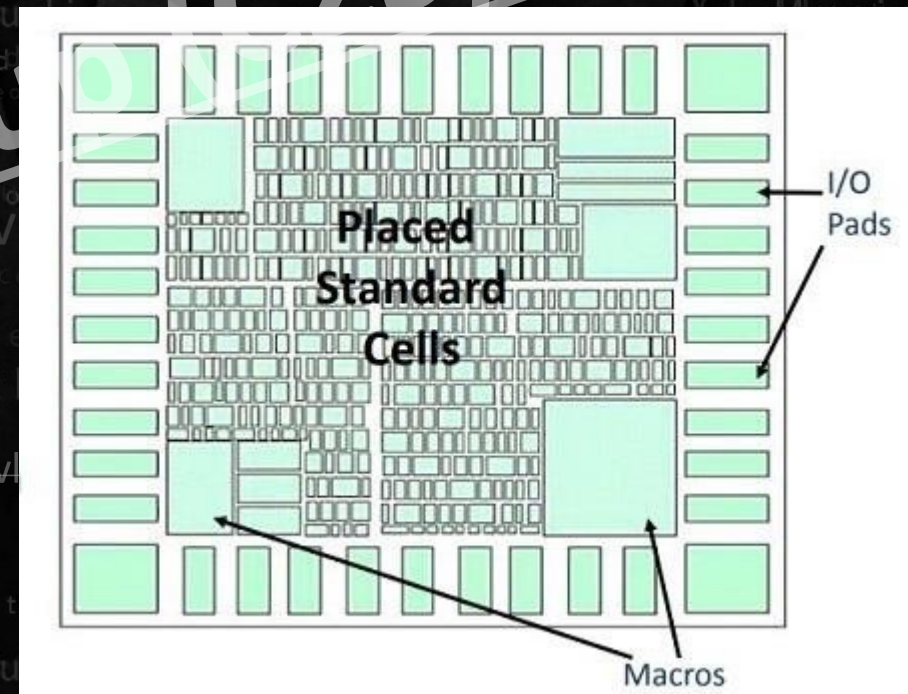
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stages involved in Physical Design.....in contrn....

c. **Placement**: process of determining a suitable physical location for each component in the design.



Initial Placement

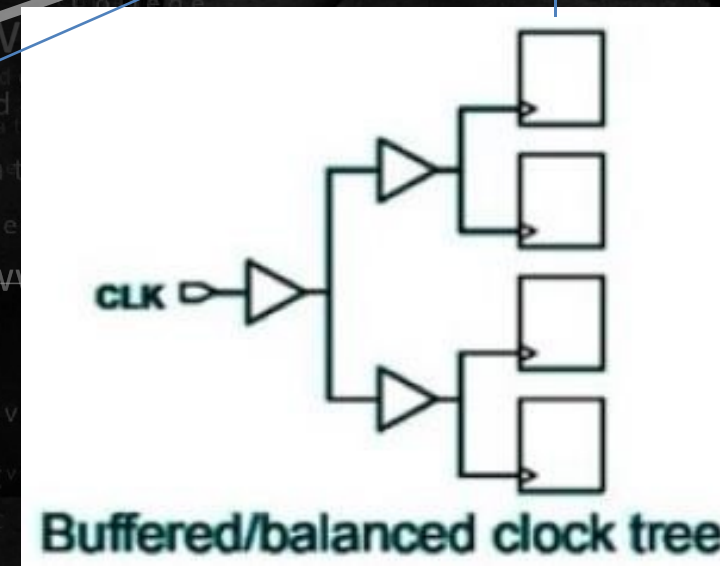
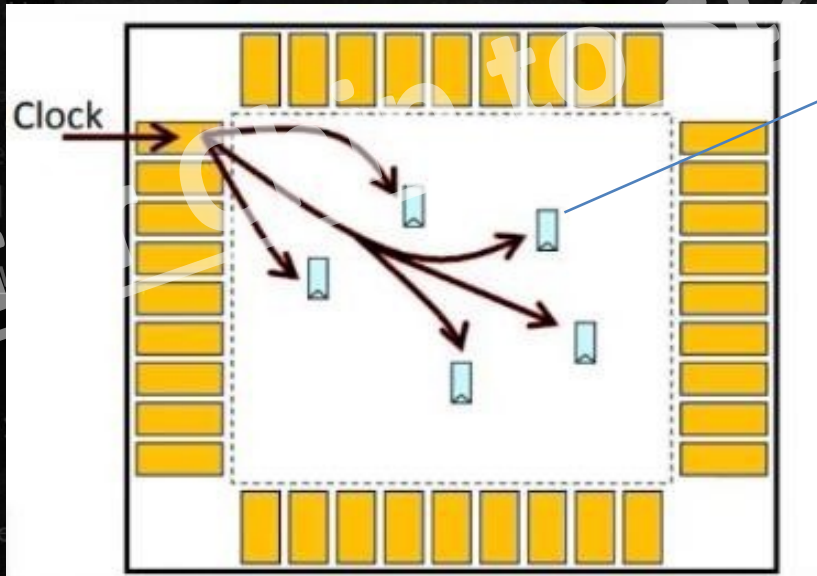


Legalized Placement

stages involved in Physical Design.....in contrn....

d. Clock Tree Synthesis (CTS)

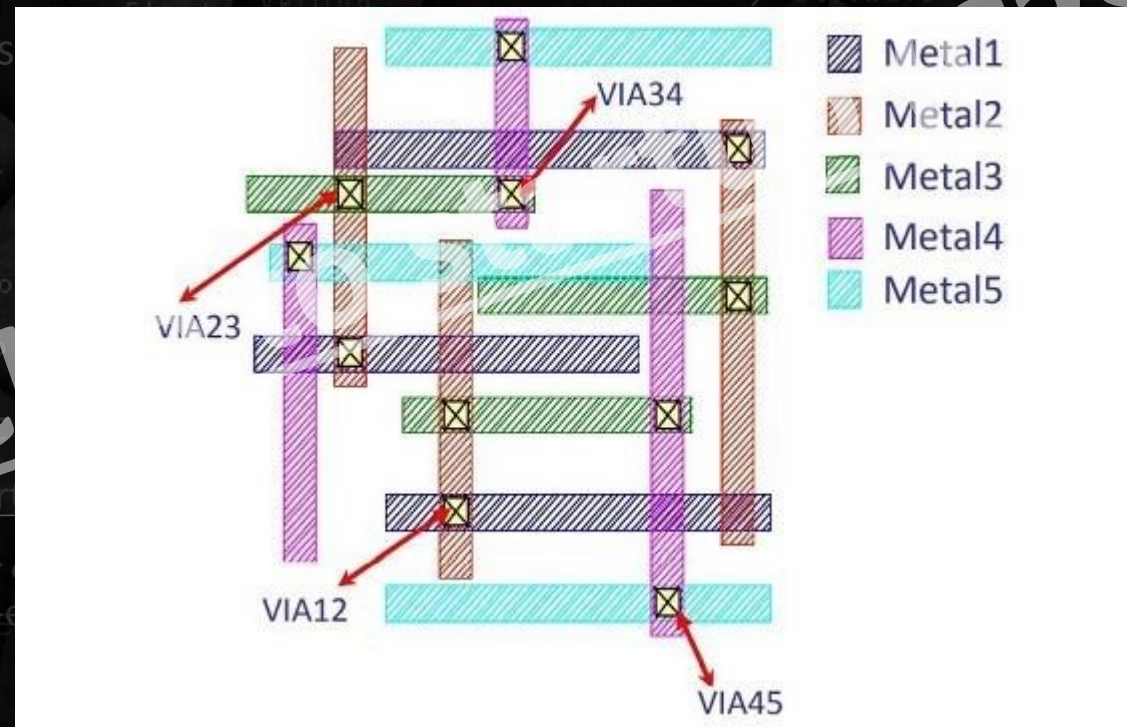
- * distributing the clock signal to all sequential elements
- * to balance the skew and latency/delay



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stages involved in Physical Design.....in contrn...

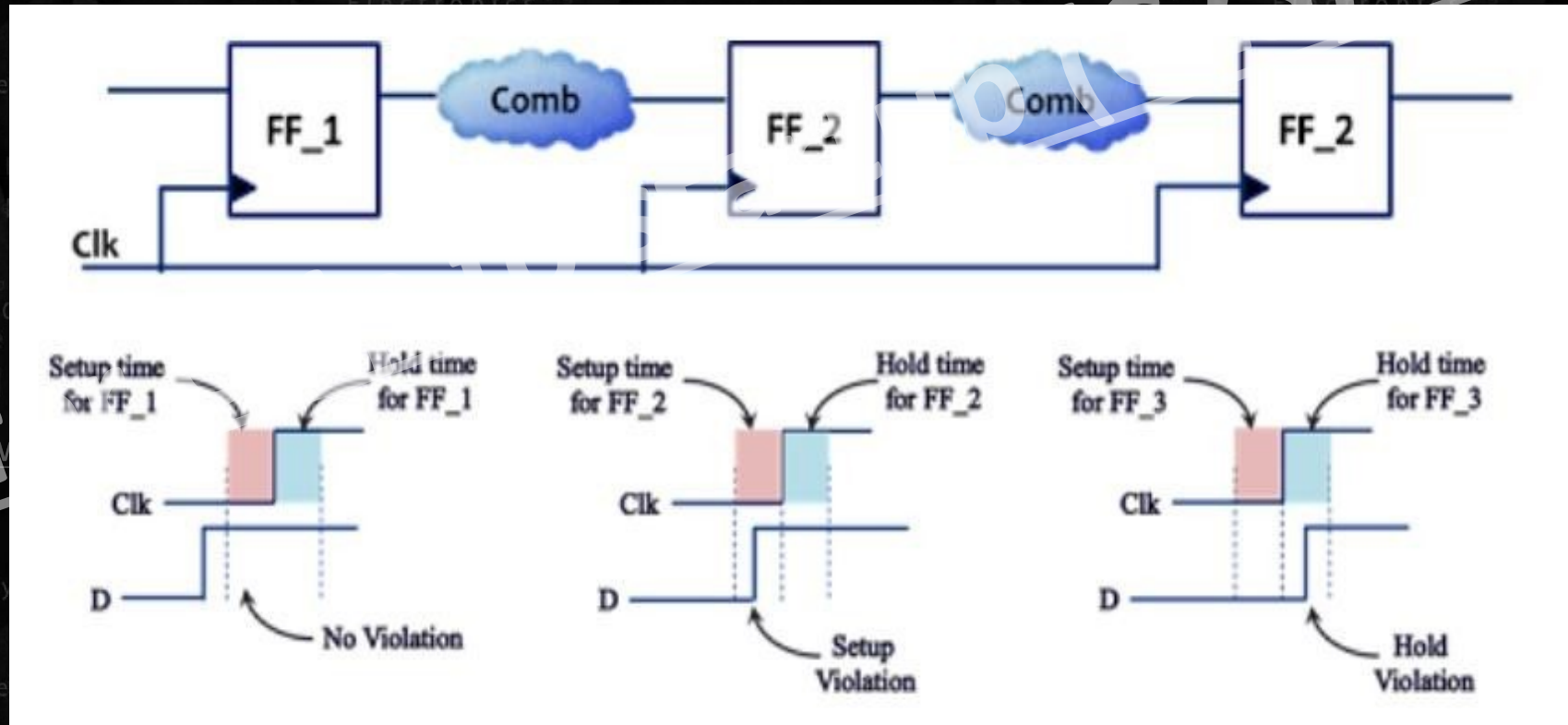
e. **Routing** – physical interconnections of standard cells, macro pins and IO pins.



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stages involved in Physical Design.....in contn...

f. Timing Analysis — to meet the timing specifications without any setup/hold violations



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stages involved in Physical Design.....in contrn...

g. Physical Verification – process to ensure zero DRC/LVS violations.

Major Checks are **DRC, LVS, ERC** and Antenna

Design Rule check (DRC) - layout satisfies a set of rules required for manufacturing

